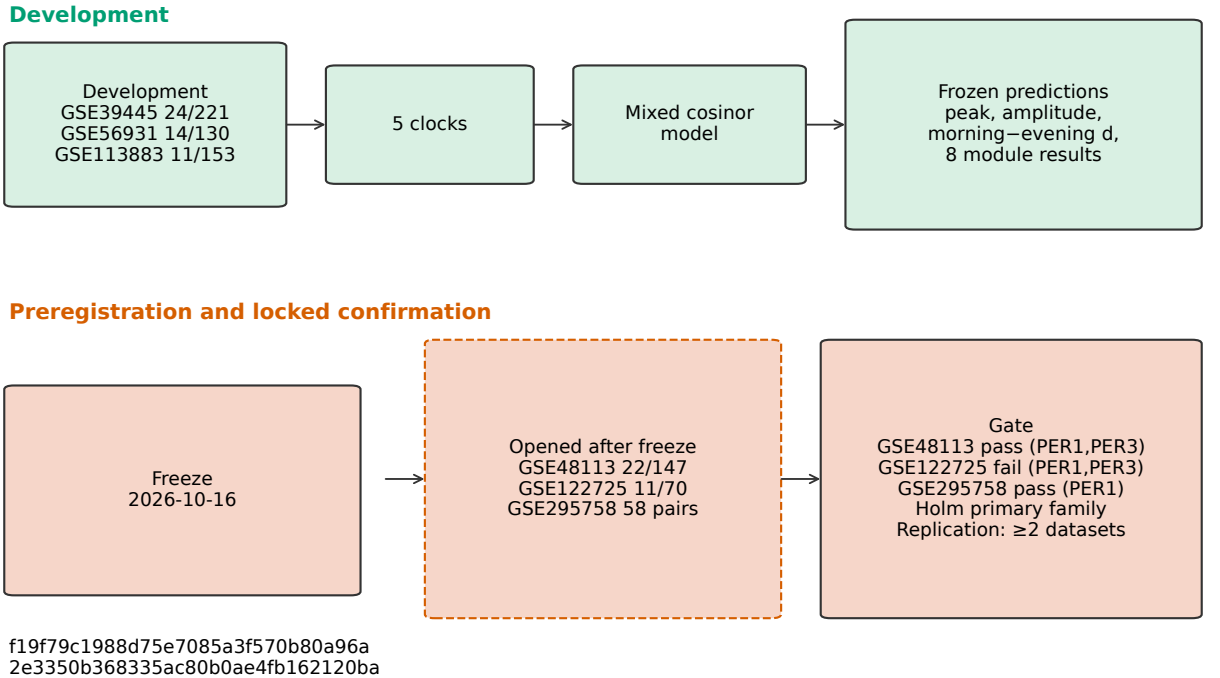
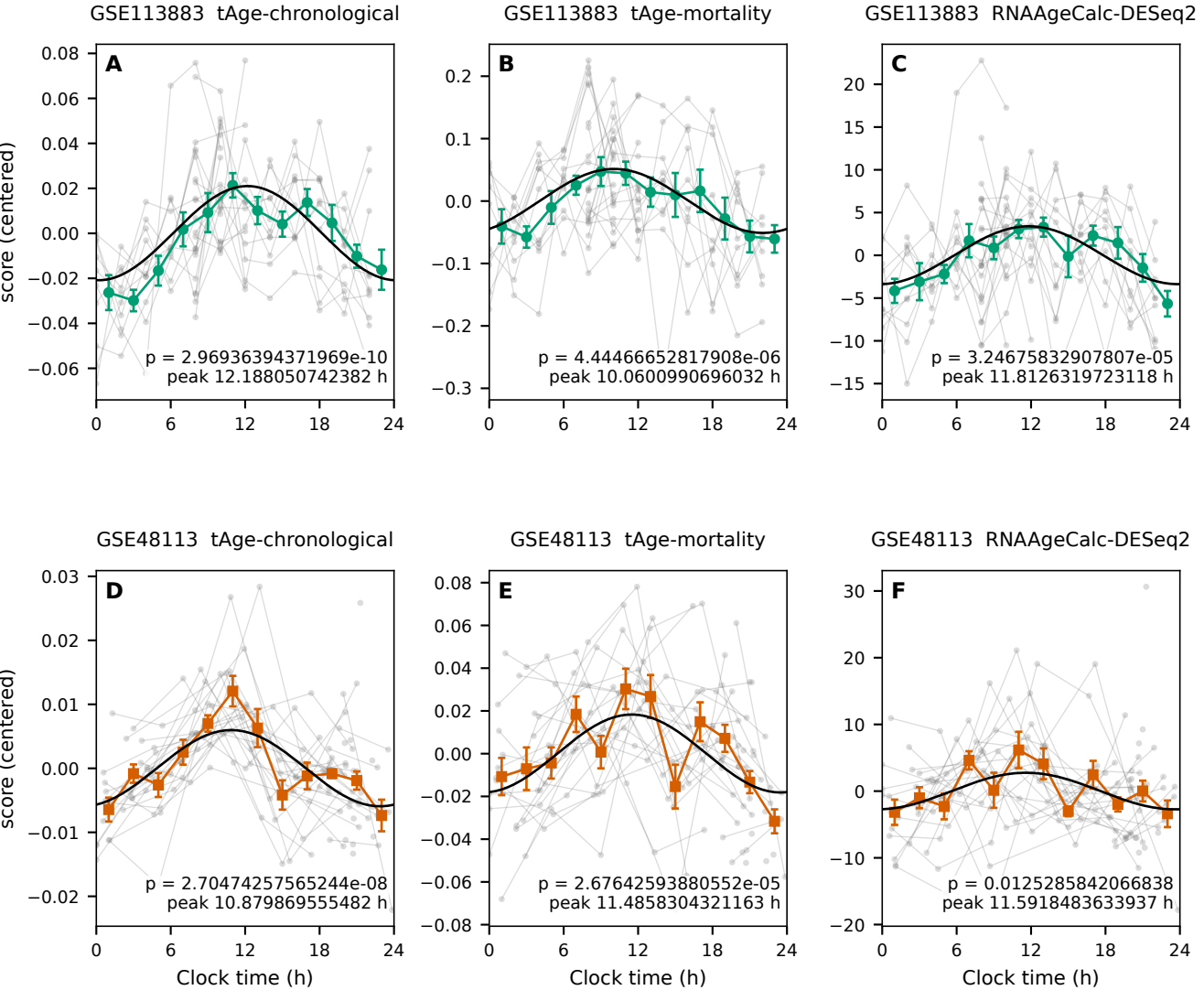
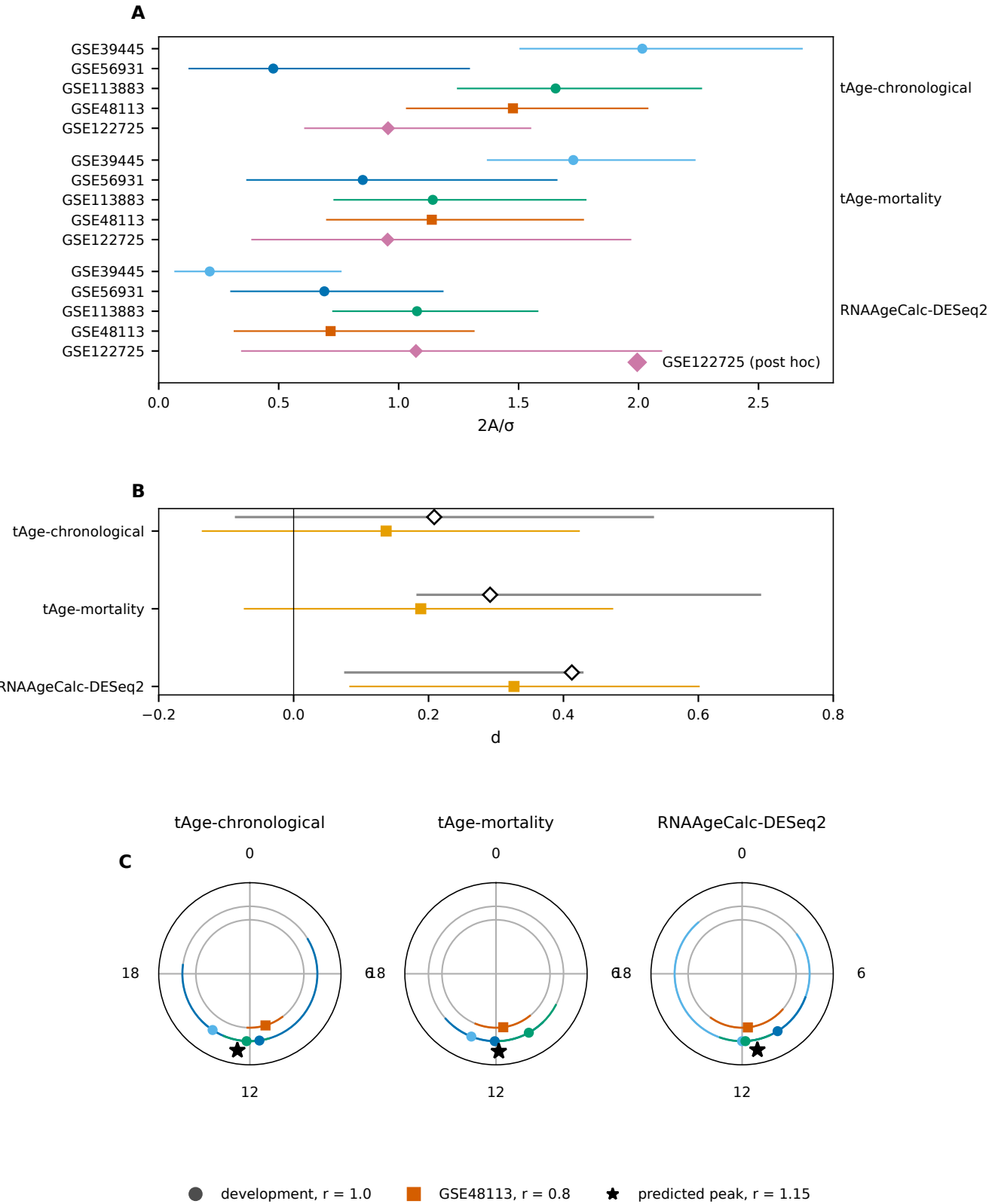
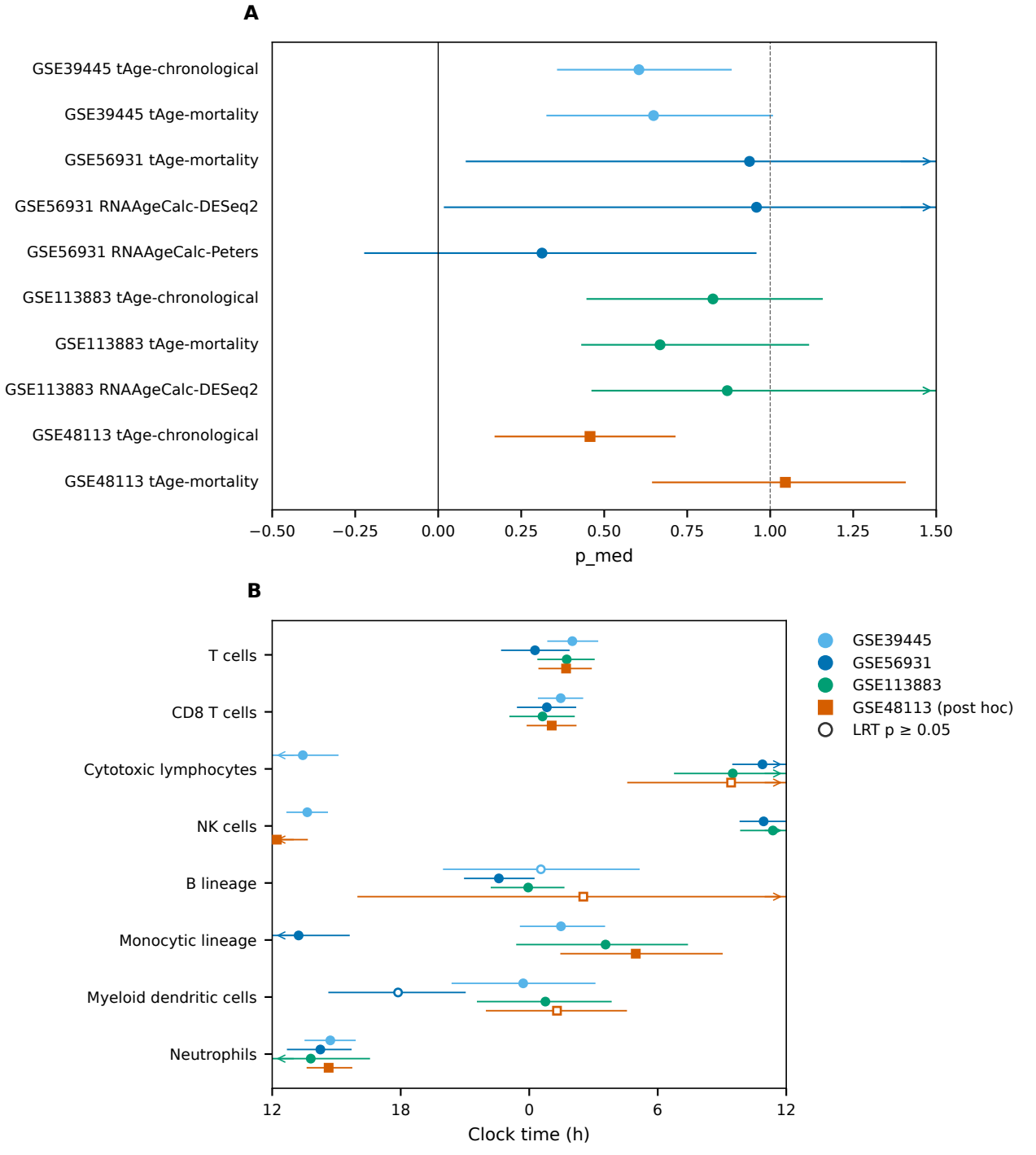
A

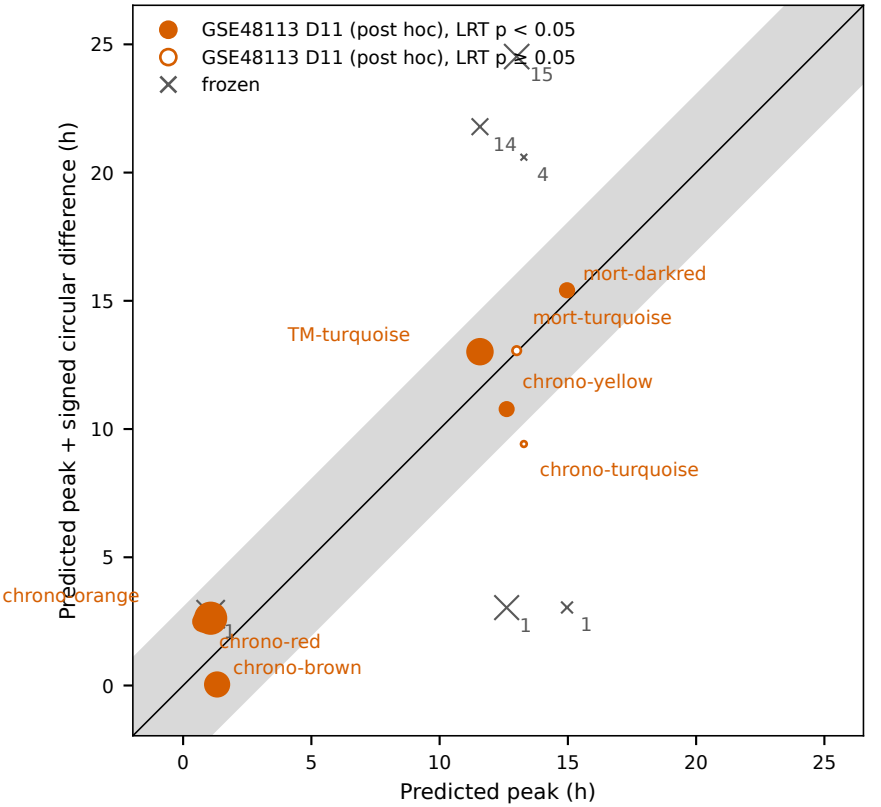

Development
Development
GSE39445 24/221
GSE56931 14/130
GSE113883 11/153
5 clocks
Mixed cosinor
model
Frozen predictions
peak, amplitude,
morning−evening d,
8 module results
Preregistration and locked confirmation
Freeze
2026-10-16
Opened after freeze
GSE48113 22/147
GSE122725 11/70
GSE295758 58 pairs
Gate
GSE48113 pass (PER1,PER3)
GSE122725 fail (PER1,PER3)
GSE295758 pass (PER1)
Holm primary family
Replication: ≥2 datasets
f19f79c1988d75e7085a3f570b80a96a
2e3350b368335ac80b0ae4fb162120ba
Post hoc
Time-axis
correction
(GSE122725)
Missing-value
sensitivity
Bias
simulation
Mitigation


GSE113883 tAge-chronological
GSE113883 tAge-mortality
GSE113883 RNAAgeCalc-DESeq2
GSE48113 tAge-chronological
GSE48113 tAge-mortality
GSE48113 RNAAgeCalc-DESeq2
A
B
C
D
E
F
score (centered)
Clock time (h)
p = 2.96936394371969e-10
peak 12.188050742382 h
p = 4.44466652817908e-06
peak 10.0600990696032 h
p = 3.24675832907807e-05
peak 11.8126319723118 h
p = 2.70474257565244e-08
peak 10.879869555482 h
p = 2.67642593880552e-05
peak 11.4858304321163 h
p = 0.0125285842066838
peak 11.5918483633937 h


A
GSE39445
GSE56931
GSE113883
GSE48113
GSE122725
GSE39445
GSE56931
GSE113883
GSE48113
GSE122725
GSE39445
GSE56931
GSE113883
GSE48113
GSE122725
GSE122725 (post hoc)
tAge-chronological
tAge-mortality
RNAAgeCalc-DESeq2
0.0
0.5
1.0
1.5
2.0
2.5
2A/σ
B
tAge-chronological
tAge-mortality
RNAAgeCalc-DESeq2
−0.2
0.0
0.2
0.4
0.6
0.8
d
C
tAge-chronological
tAge-mortality
RNAAgeCalc-DESeq2
0
6
12
18
development, r = 1.0
GSE48113, r = 0.8
predicted peak, r = 1.15

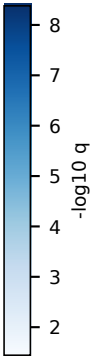

A
GSE39445 tAge-chronological
GSE39445 tAge-mortality
GSE56931 tAge-mortality
GSE56931 RNAAgeCalc-DESeq2
GSE56931 RNAAgeCalc-Peters
GSE113883 tAge-chronological
GSE113883 tAge-mortality
GSE113883 RNAAgeCalc-DESeq2
GSE48113 tAge-chronological
GSE48113 tAge-mortality
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
p_med
B
T cells
CD8 T cells
Cytotoxic lymphocytes
NK cells
B lineage
Monocytic lineage
Myeloid dendritic cells
Neutrophils
12
18
0
6
12
Clock time (h)
GSE39445
GSE56931
GSE113883
GSE48113 (post hoc)
LRT p ≥ 0.05

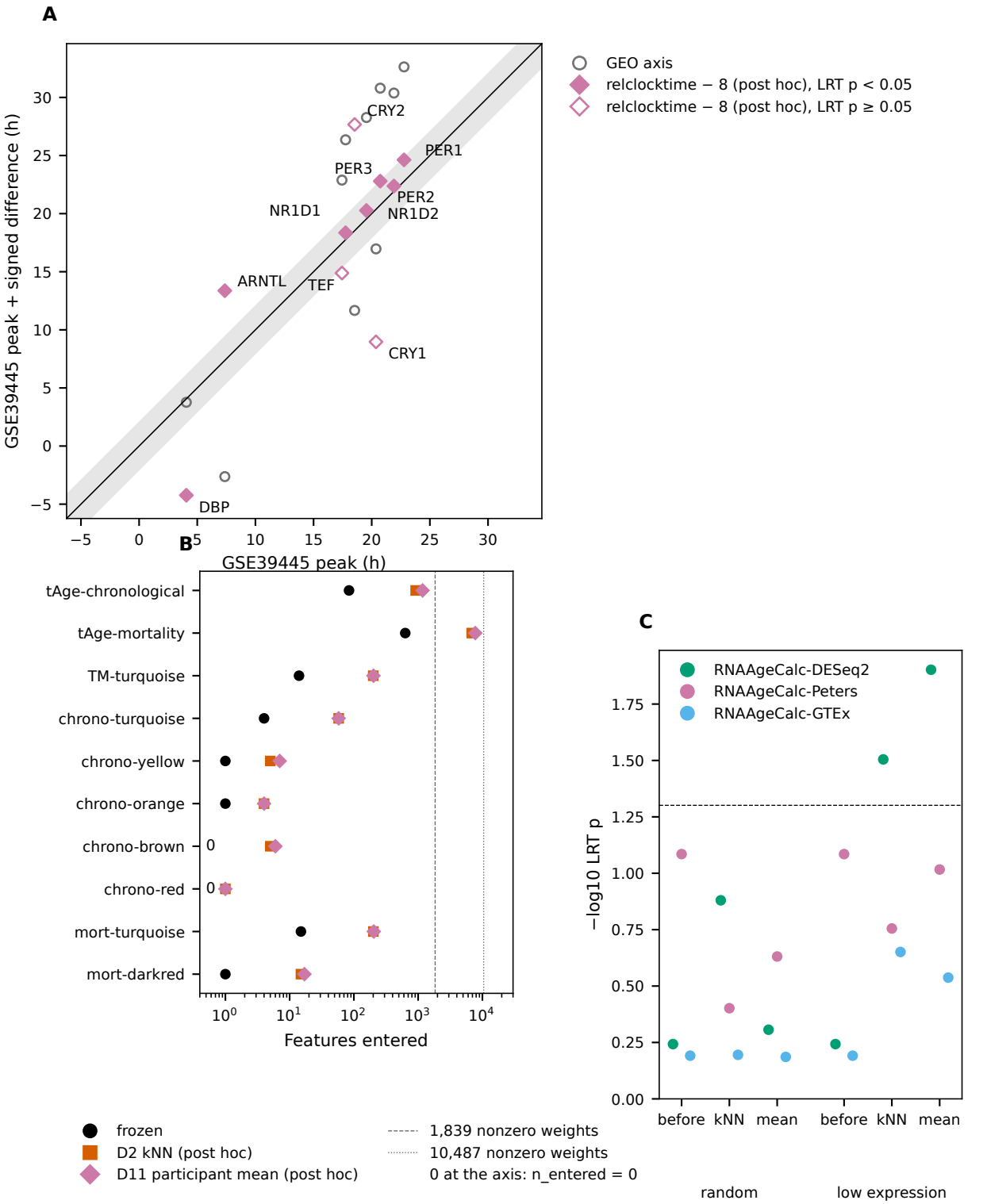
A

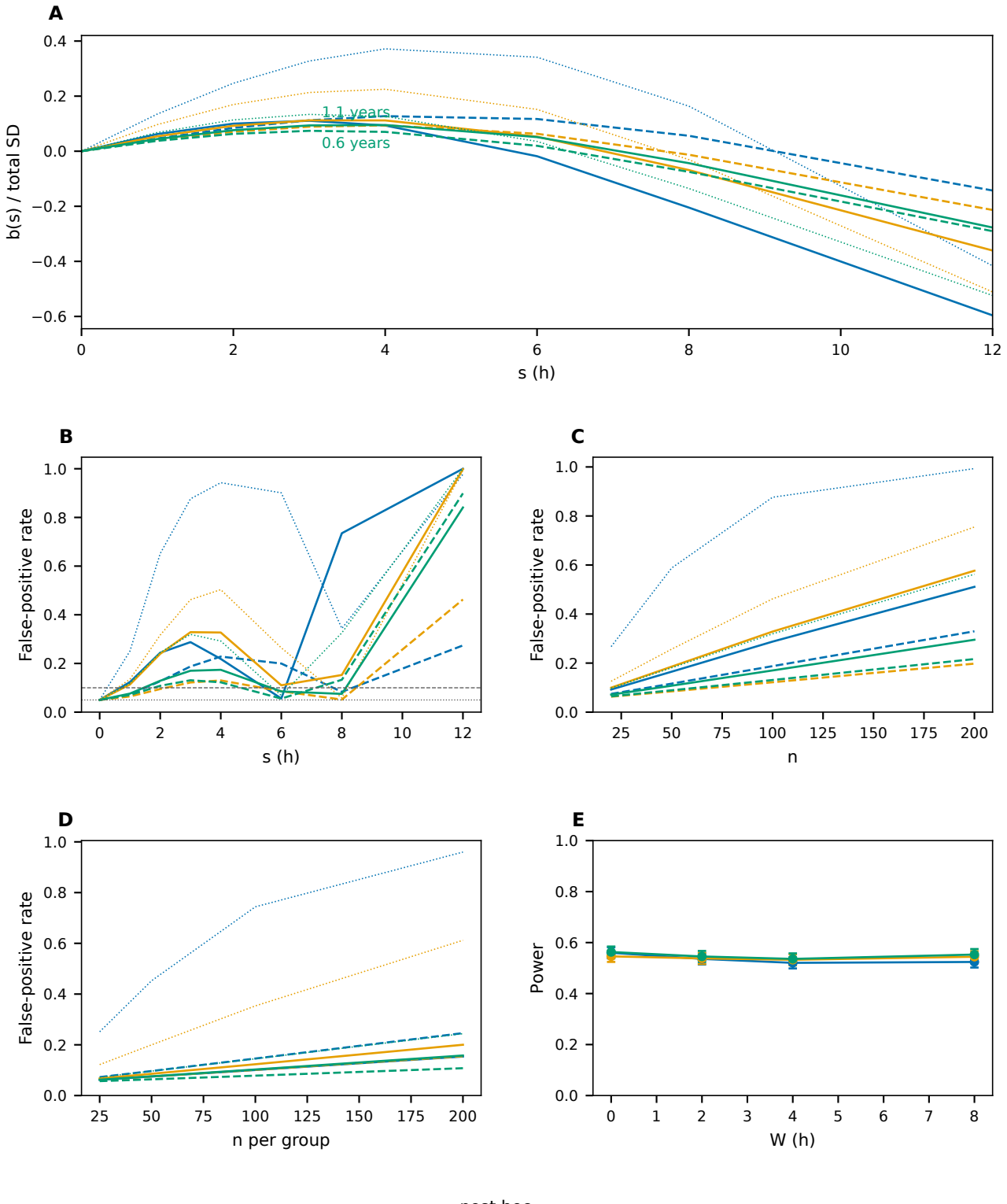


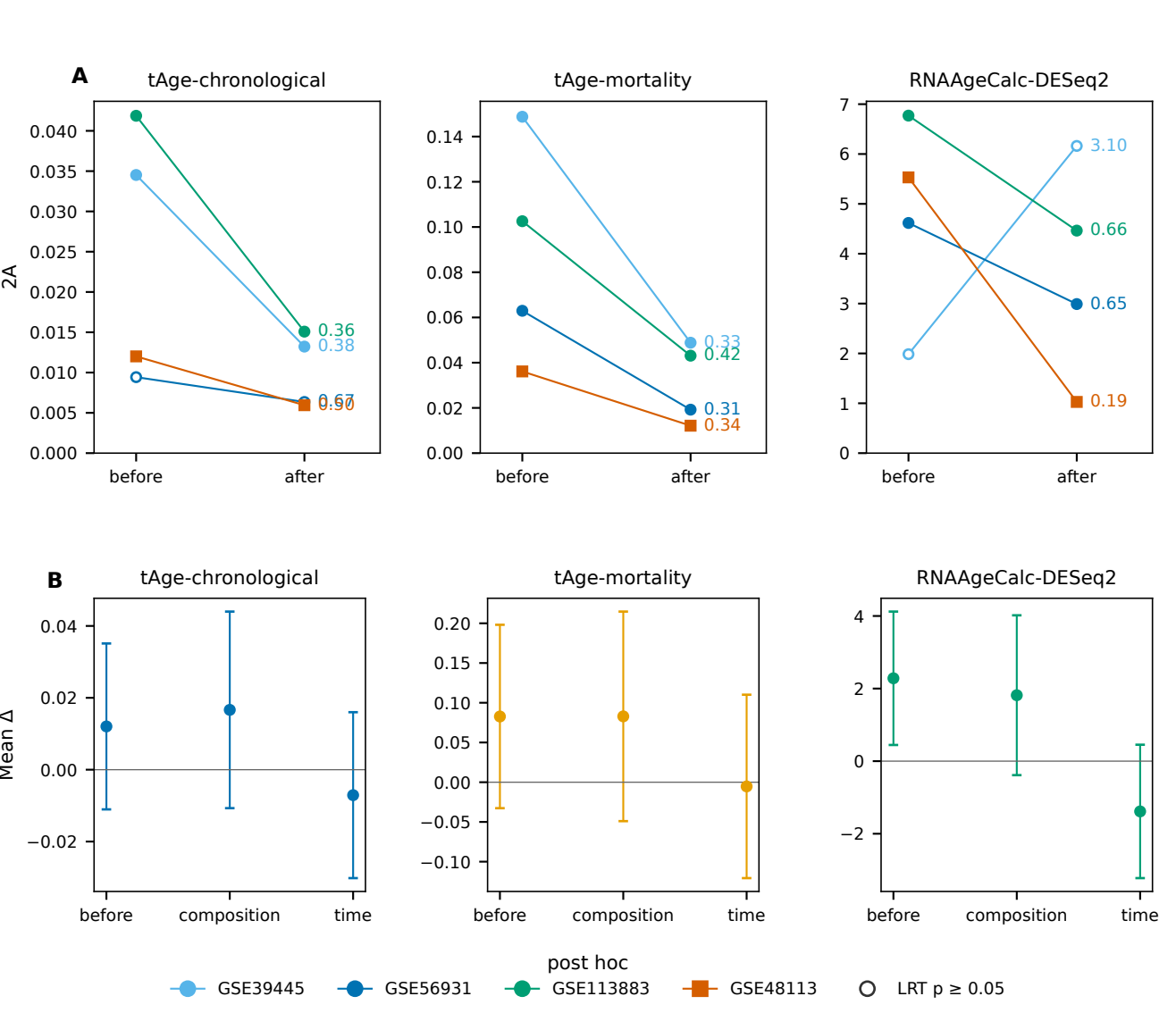
B

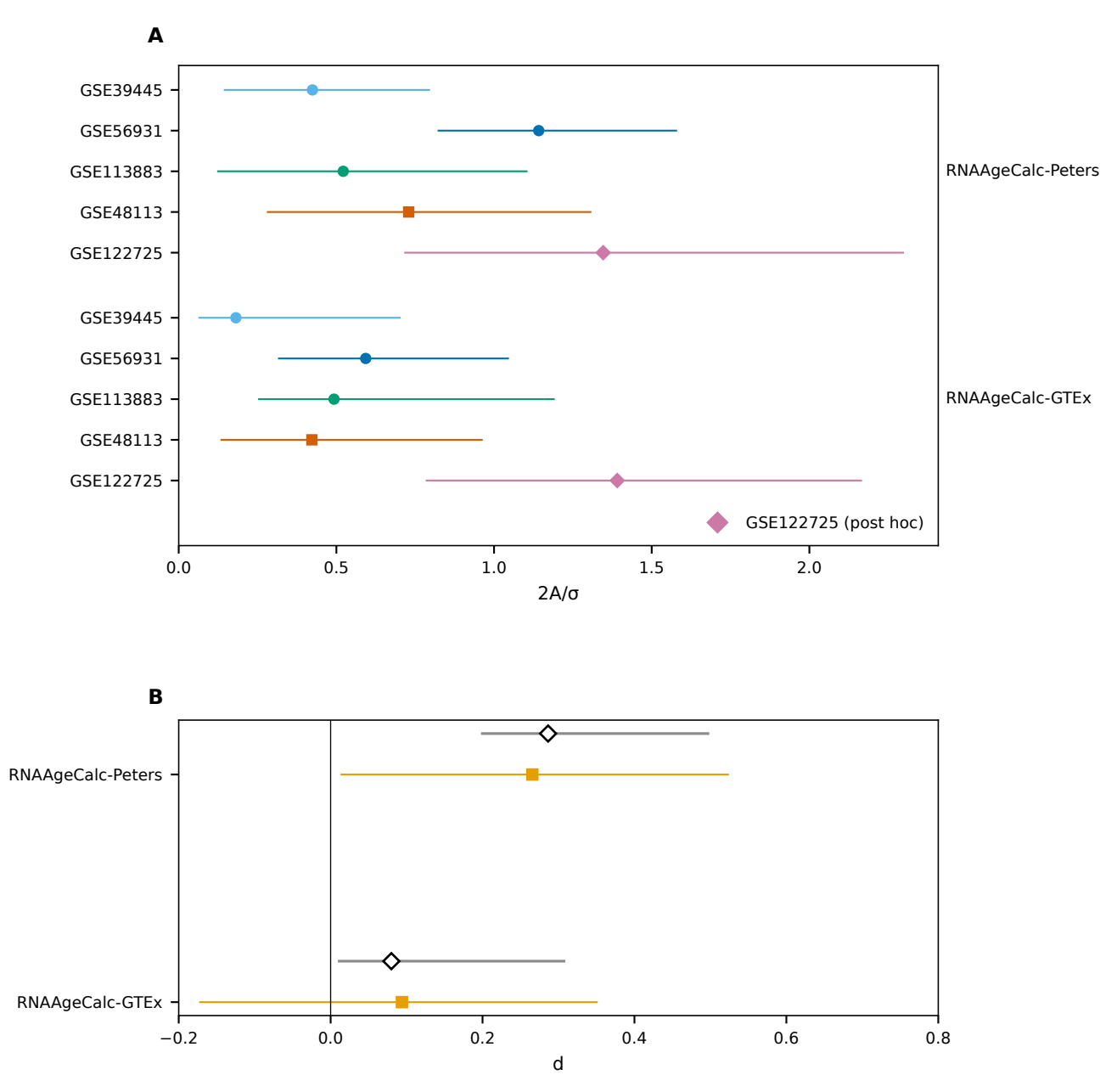
module	−log10 q 39445	−log10 q 56931	−log10 q 113883	frozen n	D11 (post hoc) n	GSE295758 pred / obs / agree
TM-turquoise				FALSE 14	TRUE 202	+ / + / TRUE
chrono-turquoise				FALSE 4	FALSE 58	+ / + / TRUE
chrono-yellow				FALSE 1	TRUE 7	+ / - /
chrono-orange				TRUE 1	TRUE 4	- / - / TRUE
chrono-brown				FALSE 0	TRUE 6	- / - /
chrono-red				FALSE 0	TRUE 1	- / + /
mort-turquoise				FALSE 15	FALSE 204	+ / + /
mort-darkred				FALSE 1	TRUE 17	- / + /

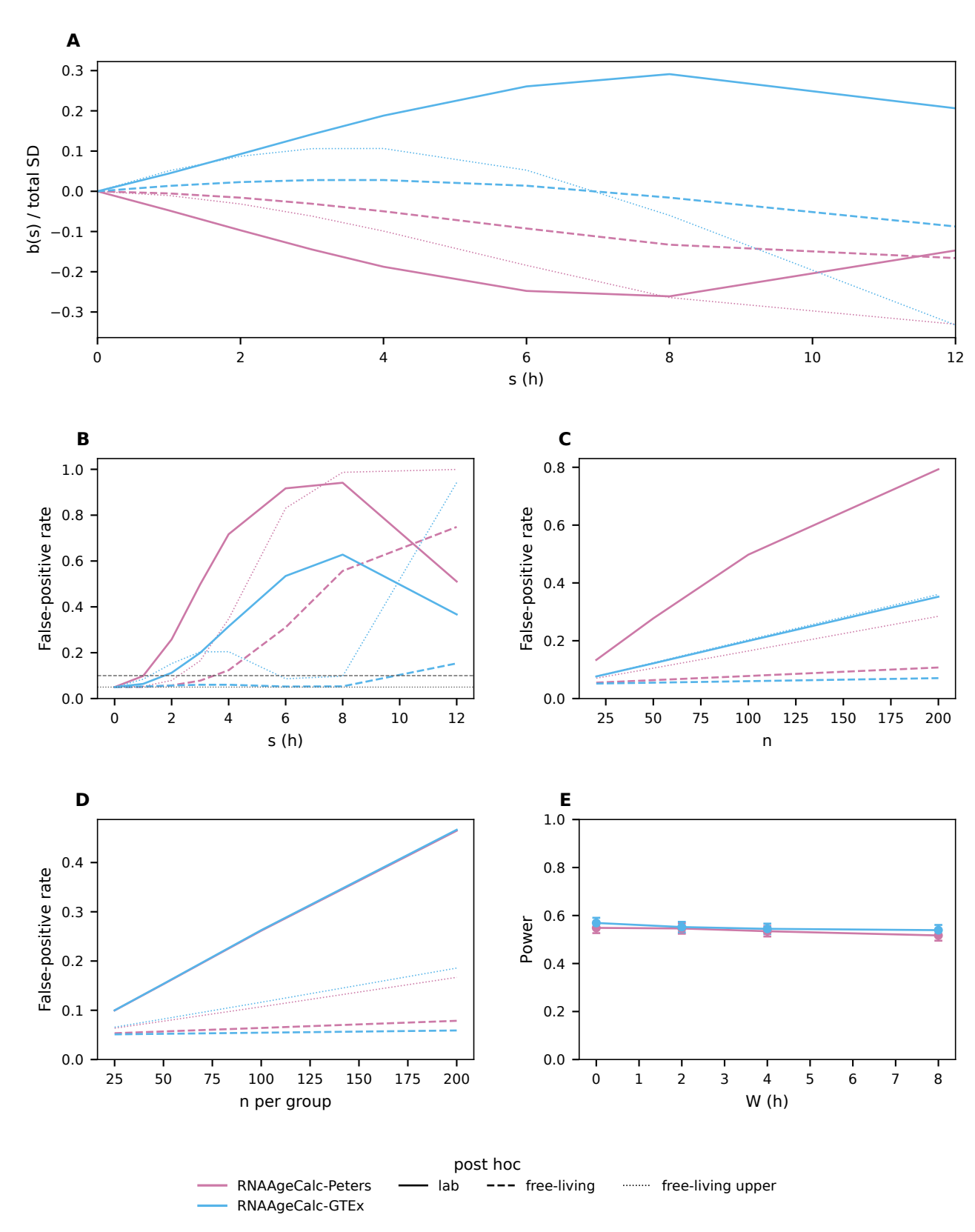


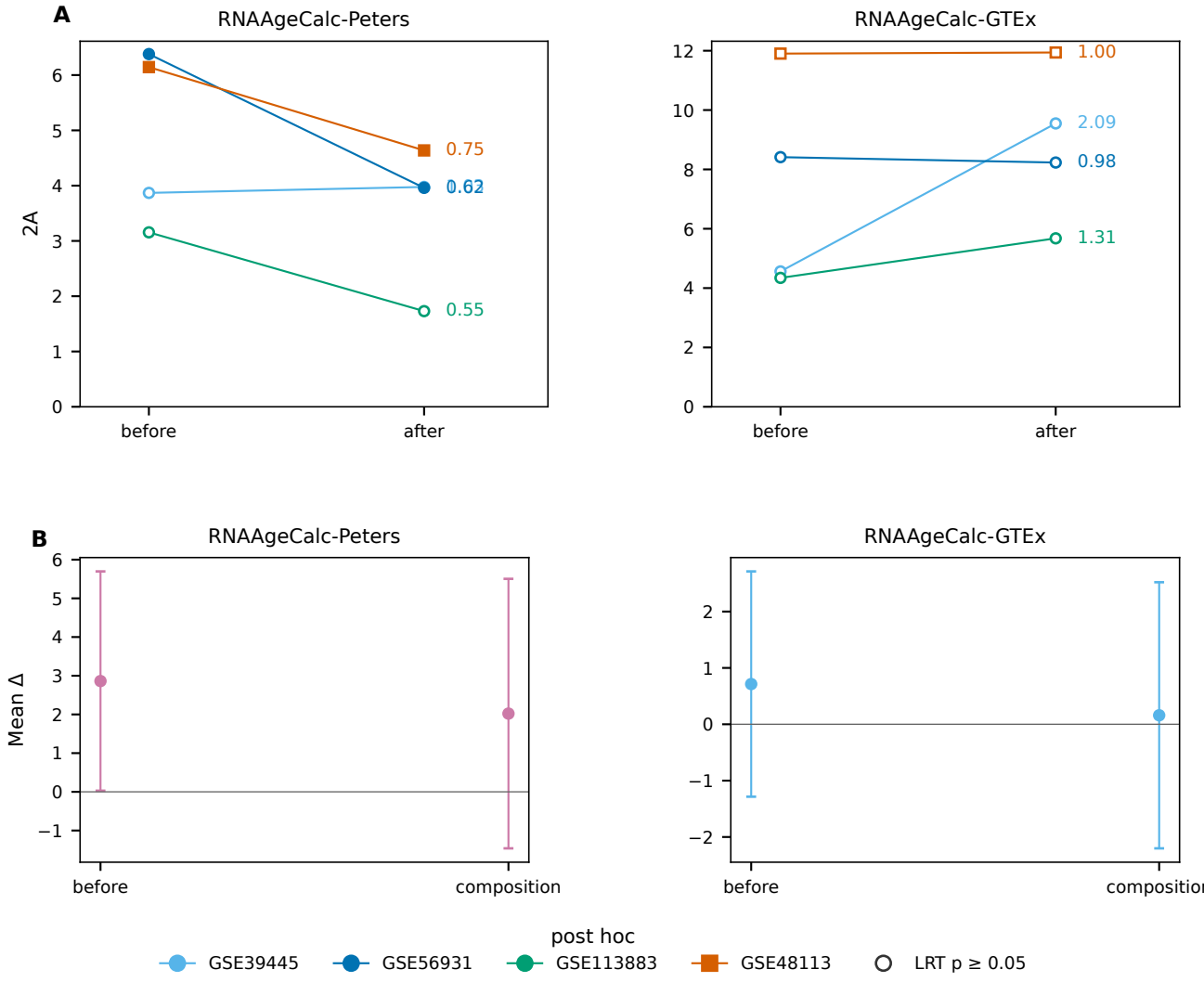

A
GSE39445 peak + signed difference (h)
GSE39445 peak (h)
CRY2
PER1
PER3
PER2
NR1D1
NR1D2
ARNTL
TEF
CRY1
DBP
GEO axis
relclocktime − 8 (post hoc), LRT p < 0.05
relclocktime − 8 (post hoc), LRT p ≥ 0.05
B
tAge-chronological
tAge-mortality
TM-turquoise
chrono-turquoise
chrono-yellow
chrono-orange
chrono-brown
chrono-red
mort-turquoise
mort-darkred
Features entered
C
RNAAgeCalc-DESeq2
RNAAgeCalc-Peters
RNAAgeCalc-GTEx
−log10 LRT p
before
kNN
mean
random
low expression
frozen
D2 kNN (post hoc)
D11 participant mean (post hoc)
1,839 nonzero weights
10,487 nonzero weights
0 at the axis: n_entered = 0


A
b(s) / total SD
s (h)
1.1 years
0.6 years
B
False-positive rate
s (h)
C
False-positive rate
n
D
False-positive rate
n per group
E
Power
W (h)
post hoc
tAge-chronological
tAge-mortality
RNAAgeCalc-DESeq2
lab
free-living
free-living upper


A
tAge-chronological
tAge-mortality
RNAAgeCalc-DESeq2
2A
0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
7
6
5
4
3
2
1
0
0.36
0.38
0.67
0.90
0.33
0.42
0.31
0.34
3.10
0.66
0.65
0.19
before
after
B
Mean Δ
0.04
0.02
0.00
−0.02
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
4
2
0
−2
before
composition
time
post hoc
GSE39445
GSE56931
GSE113883
GSE48113
LRT p ≥ 0.05


A
GSE39445
GSE56931
GSE113883
GSE48113
GSE122725
RNAAgeCalc-Peters
GSE39445
GSE56931
GSE113883
GSE48113
GSE122725
RNAAgeCalc-GTEx
GSE122725 (post hoc)
0.0
0.5
1.0
1.5
2.0
2A/σ
B
RNAAgeCalc-Peters
RNAAgeCalc-GTEx
−0.2
0.0
0.2
0.4
0.6
0.8
d


A
b(s) / total SD
s (h)
B
False-positive rate
s (h)
C
False-positive rate
n
D
False-positive rate
n per group
E
Power
W (h)
post hoc
RNAAgeCalc-Peters
RNAAgeCalc-GTEx
lab
free-living
free-living upper


A
RNAAgeCalc-Peters
2A
0.75
0.62
0.55
before
after
RNAAgeCalc-GTEx
1.00
2.09
0.98
1.31
before
after
B
RNAAgeCalc-Peters
Mean Δ
before
composition
RNAAgeCalc-GTEx
before
composition
post hoc
GSE39445
GSE56931
GSE113883
GSE48113
LRT p ≥ 0.05